Web Programming

Lab Assignment 14

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LAB 14 – HTML Code:

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Detach Event Listener</title>
        <link rel="stylesheet" href="styles.css">
    </head>
    <body>
        <h1>Detach Event Listener</h1>
        <p>This page shows how an event listener can be attached to an element and then
removed.</p>
        <button id="actionBtn">Click To Demonstrate Listener Status</button>
        <button id="listenerBtn">Click To Attach Listener</button>
        <div id="log"></div>
        <script src="script.js"></script>
    </body>
</html>
```

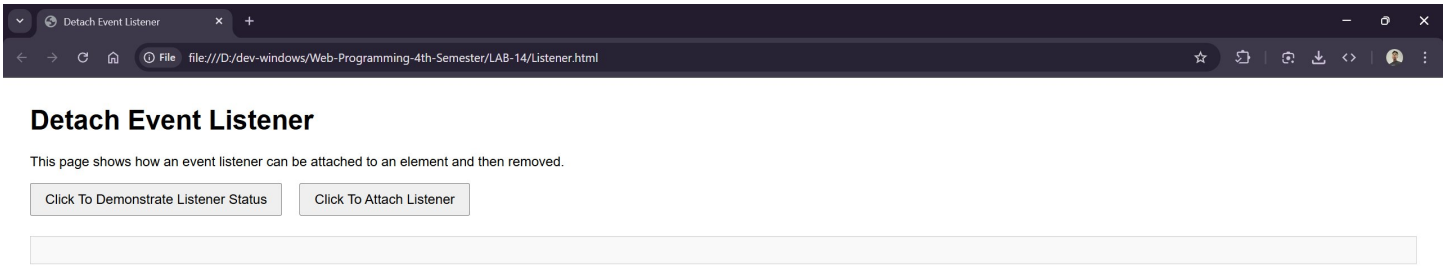
LAB 14 – CSS Code:

```
body {
    font-family: Arial, sans-serif;
    margin: 2rem;
}

button {
    margin-right: 1rem;
    padding: 0.5rem 1rem;
    font-size: 1rem;
    cursor: pointer;
}

#log {
    margin-top: 1.5rem;
    border: 1px solid #ccc;
    padding: 1rem;
    max-height: 200px;
    overflow-y: auto;
    background: #f9f9f9;
}

#log p {
    margin: 0.25rem 0;
}
```

LAB 14 – JavaScript Code:

```
// Attach an event listener to action button
const actionBtn = document.getElementById('actionBtn');
const listenerBtn = document.getElementById('listenerBtn');
const logDiv = document.getElementById('log');

function handleActionClick() {
    const msg = 'Button clicked at ' + new Date().toLocaleTimeString();
    const p = document.createElement('p');
    p.textContent = msg;
    logDiv.appendChild(p);
}

function handleListenerClickAttach() {
    actionBtn.addEventListener('click', handleActionClick);
    listenerBtn.textContent = "Click To Detach Listener";
    const p = document.createElement('p');
    p.textContent = 'Listener attached';
    p.style.fontStyle = 'italic';
    logDiv.appendChild(p);
}

function handleListenerClickDetach() {
    actionBtn.removeEventListener('click', handleActionClick);
    listenerBtn.textContent = "Click To Attach Listener";
    const p = document.createElement('p');
    p.textContent = 'Listener detached';
    p.style.fontStyle = 'italic';
    logDiv.appendChild(p);
}

var state = 0;

// When the listener button is initially pressed we attach the listener, when pressed again
we detach the listener, and so on and so forth. We also log the current state of the listener
in the log div.
listenerBtn.addEventListener('click', () => {
    if (state == 0) {
        handleListenerClickAttach();
        state = 1;
    }
    else {
        handleListenerClickDetach();
        state = 0;
    }
});
```

LAB 14 – Screenshots:



Detach Event Listener

This page shows how an event listener can be attached to an element and then removed.

Click To Demonstrate Listener Status | Click To Detach Listener

Listener attached

Detach Event Listener

This page shows how an event listener can be attached to an element and then removed.

Click To Demonstrate Listener Status | Click To Detach Listener

Listener attached
Button clicked at 2:48:34 PM
Button clicked at 2:48:36 PM
Button clicked at 2:48:36 PM

Detach Event Listener

This page shows how an event listener can be attached to an element and then removed.

Click To Demonstrate Listener Status | Click To Attach Listener

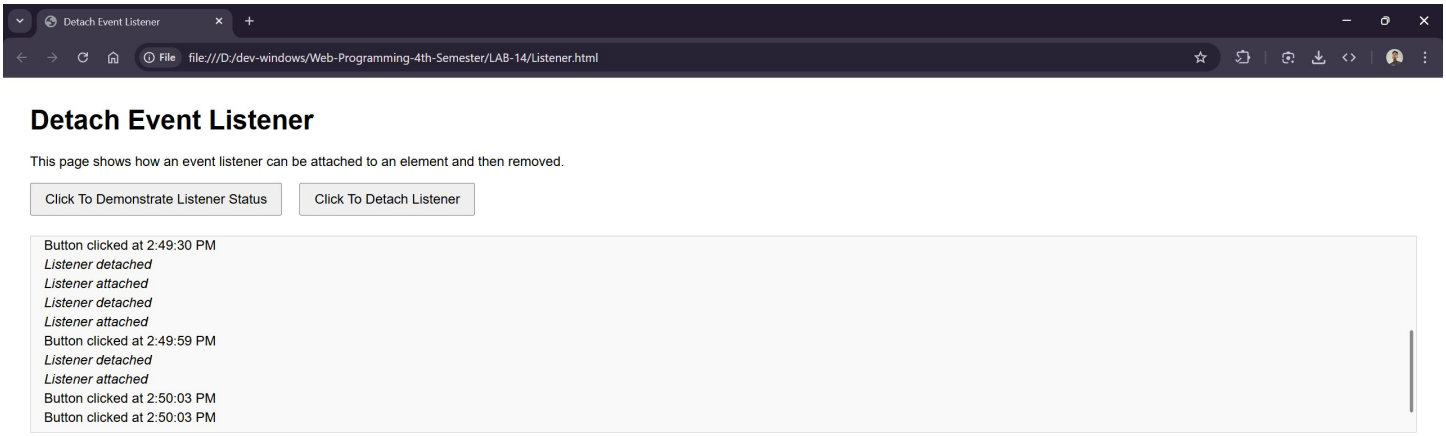
Listener attached
Button clicked at 2:48:34 PM
Button clicked at 2:48:36 PM
Button clicked at 2:48:36 PM
Button clicked at 2:49:06 PM
Listener detached

Detach Event Listener

This page shows how an event listener can be attached to an element and then removed.

Click To Demonstrate Listener Status | Click To Detach Listener

Listener attached
Button clicked at 2:48:34 PM
Button clicked at 2:48:36 PM
Button clicked at 2:48:36 PM
Button clicked at 2:49:06 PM
Listener detached
Listener attached
Button clicked at 2:49:25 PM
Button clicked at 2:49:26 PM
Button clicked at 2:49:30 PM


Detach Event Listener
file:///D:/dev-windows/Web-Programming-4th-Semester/LAB-14/Listener.html
Detach Event Listener
This page shows how an event listener can be attached to an element and then removed.
Click To Demonstrate Listener Status
Click To Detach Listener
Button clicked at 2:49:30 PM
Listener detached
Listener attached
Listener detached
Listener attached
Button clicked at 2:49:59 PM
Listener detached
Listener attached
Button clicked at 2:50:03 PM
Button clicked at 2:50:03 PM

THANK YOU!!!